HotelRevAI

AI-Driven Revenue Analysis for Hotels

Submitted by:

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Abstract

This report presents a comprehensive data-driven analysis of hotel booking performance using Power BI. The HotelRevAI project leverages a real-world hotel bookings dataset to identify revenue trends, customer behaviour patterns, and actionable business insights. The project encompasses the complete data analytics pipeline — from data ingestion and preprocessing, through structured data modelling and advanced DAX measure development, to the creation of four interactive dashboards with rich visual storytelling.

The dashboards deliver insights across four analytical dimensions: overall revenue performance, customer and booking behaviour, revenue optimization with AI-derived insights, and seasonality and stay pattern analysis. By transforming raw transactional data into interactive intelligence, the project demonstrates how data visualization can empower hotel management to make informed, strategic decisions that optimize revenue and enhance guest experience.

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1. Introduction

The hospitality industry is characterized by high-volume transactional data, seasonal demand fluctuations, and complex customer segmentation, all of which demand robust analytical tools to extract meaningful insights. HotelRevAI is a Power BI-based analytics project designed to address this need by providing hotel managers and revenue analysts with interactive dashboards and data-driven KPIs.

This project uses hotel booking data sourced from Kaggle, which captures booking records for both a City Hotel and a Resort Hotel. The dataset includes over 75,000 bookings spanning information such as arrival dates, length of stay, customer types, distribution channels, Average Daily Rate (ADR), and cancellation status. Through structured data loading, transformation, DAX modelling, and Power BI visualization, the project transforms this raw data into actionable business intelligence.

1.1 Problem Statement

Hotels face significant challenges in understanding revenue fluctuations, booking patterns, and customer segments due to the large volumes of raw booking data generated daily. Without proper visualization and analytical frameworks, it is difficult for hotel management to:

- Identify peak demand seasons and optimize pricing accordingly
- Understand which distribution channels and customer types drive the most revenue
- Evaluate the financial impact of cancellations on overall performance
- Track customer loyalty and repeated booking patterns
- Make data-driven decisions around occupancy management and stay duration strategies

HotelRevAI addresses these challenges by building a complete Power BI analytics solution — from data model to visual dashboard — that provides hotel stakeholders with clear, interactive, and actionable insights.

1.2 Project Objectives

- Analyse overall revenue performance and identify growth trends over time.
- Evaluate hotel KPIs including ADR, cancellation rate, total bookings, and guest counts.
- Analyse customer behaviour and booking patterns by customer type and distribution channel.
- Develop DAX measures and a structured Business Matrix framework for all analytical dimensions.

- Create interactive Power BI dashboards with dynamic filtering by year, hotel type, and customer type.
- Enable data-driven decision-making in pricing, occupancy management, and customer loyalty strategy.

2. Dataset Overview

This is a hotel booking dataset collected from Kaggle. It contains booking information for two hotel types — a City Hotel and a Resort Hotel — and covers multiple years of transactional data. This includes information such as when the booking was made, length of stay, the number of adults, children, and/or babies, and the number of available parking spaces, among other things.

3. Data Preprocessing

The dataset was imported into Power BI from a CSV file containing hotel booking records. Power BI's Power Query Editor was used to preview, transform, and prepare the data before loading it into the data model. The following preprocessing steps were performed to ensure data quality and analytical accuracy:

3.1 Data Cleaning Operations

- Handled missing or null values in critical columns including country, agent, and company fields
- Verified and corrected data types — date, numeric, and text columns were standardised
- Removed duplicate records to prevent double-counting in aggregated measures
- Removed columns not participating in revenue analysis to improve model performance

3.2 Derived Columns Created

- Total Stay Duration: calculated as the sum of weekend nights and weekday nights
- Total Guests: sum of adults, children, and babies per booking
- Revenue: calculated using ADR multiplied by total nights stayed
- Arrival Date: a consolidated date field combining arrival year, month, and day-of-month columns

4. Data Modelling

A structured relational data model was developed in Power BI to support multi-dimensional analysis across all dashboards. The model follows a star schema design with a central fact table connected to dimension tables via defined relationships.

4.1 Model Components

Fact Table: Hotel_Data

The core transactional table containing booking-level data including customer type, ADR, reserved room type, booking changes, country, cancellation status, and arrival date. This table forms the basis for all revenue and booking measures.

Dimension Table (Calendar):

This table contains transactional booking-level data such as customer type, ADR, reserved room type, booking changes, country, and arrival date. This table includes:

- Date
- Month
- Month Number
- Year

Measures Table: d3_measures

A dedicated measures table was created to store all calculated DAX metrics, maintaining a clean separation between raw data and computed analytics. This table includes 16 business metrics spanning revenue, bookings, cancellations, and guest behaviour.

4.2 Model Relationship

A one-to-many relationship was established between the Calendar dimension and the fact table:

Calendar[Date] → Hotel_data[Arrival Date] (One-to-Many, Single Cross-filter Direction)

This relationship enables all DAX time-based measures to correctly aggregate data when a year or month filter is applied from any slicer or visual across the dashboards.

5. DAX Measure Library

DAX (Data Analysis Expressions) is the formula language used by Power BI to create custom calculations. In HotelRevAI, all measures are stored in the d3_measures table and operate under two evaluation contexts: Row Context (applied row-by-row) and Filter Context (applied based on active slicers and relationships). These measures include:

- Total Revenue
- Total Customers
- Total Guests Count
- Total Nights Stayed
- Revenue per Night
- Cancelled Bookings
- Cancellation Rate
- Revenue Loss
- Cancellation Impact
- Bookings New
- Bookings Repeated
- Repeated Guests
- Repeat %
- Occupancy Proxy
- Revenue Repeated

6. Business Matrix Framework

The Business Matrix maps every DAX measure to a strategic business dimension, ensuring each metric serves a defined decision-making purpose aligned with hotel management goals. Five business domains are covered:

Financial Performance

- Total Revenue → Track overall revenue & set budget targets
- Revenue per Night → Evaluate nightly pricing vs ADR
- Revenue Loss → Quantify cancellation financial impact

Customer Behaviour

- Total Customers → Understand demand volume & acquisition trends
- Repeat % → Gauge loyalty program effectiveness
- Repeated Guests → Target loyal customers for upselling

Occupancy & Demand

- Total Guests Count → Monitor occupancy demand & room utilisation
- Occupancy Proxy → Inform room allocation strategies
- Total Nights Stayed → Analyse stay duration for extended-stay pricing

Booking Patterns

- Total Bookings → Track volumes & identify seasonal peaks
- Bookings New → Measure new customer acquisition rate
- Bookings Repeated → Track loyalty return rate vs new acquisition

7. Dashboard Visualization & Storytelling

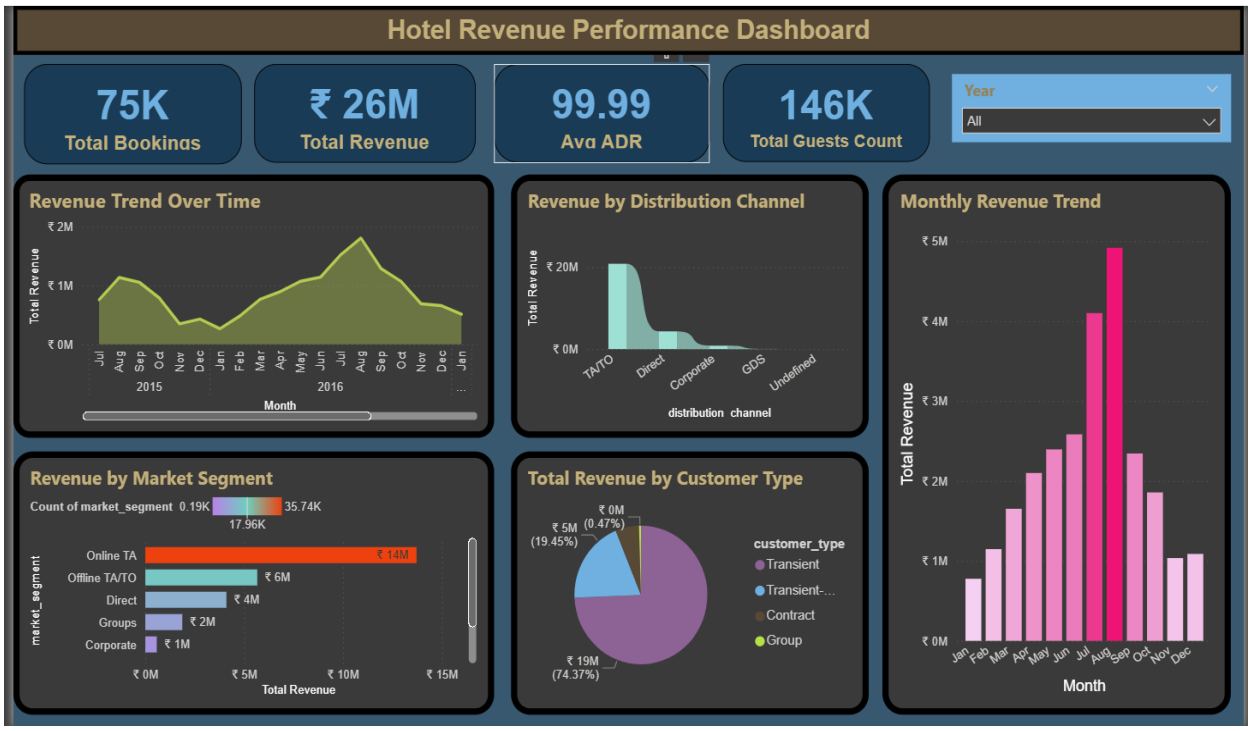
Four interactive dashboards were developed in Power BI to present the hotel data across different analytical lenses. Each dashboard is designed with a specific narrative focus, enabling stakeholders to explore insights interactively through slicers, filters, and drill-through capabilities.

7.1 Hotel Revenue Performance Dashboard

This dashboard provides a consolidated executive overview of hotel performance through high-level KPIs and trend visualizations. It is the primary entry point for understanding the overall financial health of the hotel portfolio.

Key Visualizations

- **Revenue Trend Over Time:** Line chart displaying monthly revenue across 2015–2017, revealing seasonal peaks in mid-year months
- **Revenue by Distribution Channel:** Area chart showing TA/TO channels dominating at ₹20M+, far ahead of Direct and Corporate channels
- **Revenue by Market Segment:** Horizontal bar chart with Online TA generating the highest revenue at ₹14M, followed by Offline TA/TO at ₹6M
- **Total Revenue by Customer Type:** Pie chart showing Transient customers contributing 74.37% of total revenue
- **Monthly Revenue Trend:** Column chart revealing August as the peak revenue month with nearly ₹5M generated



Business Story & Insights

- Online travel agents are the dominant booking channel, representing the highest revenue stream — signalling the importance of maintaining competitive OTA listings and rates
- Transient customers drive three-quarters of all revenue, indicating an opportunity to develop targeted offers and loyalty programmes for this segment
- August shows a consistent revenue spike across years, creating a predictable peak season that management should leverage with dynamic pricing strategies
- Revenue distribution across channels suggests underutilisation of direct booking and corporate segments — areas for strategic growth

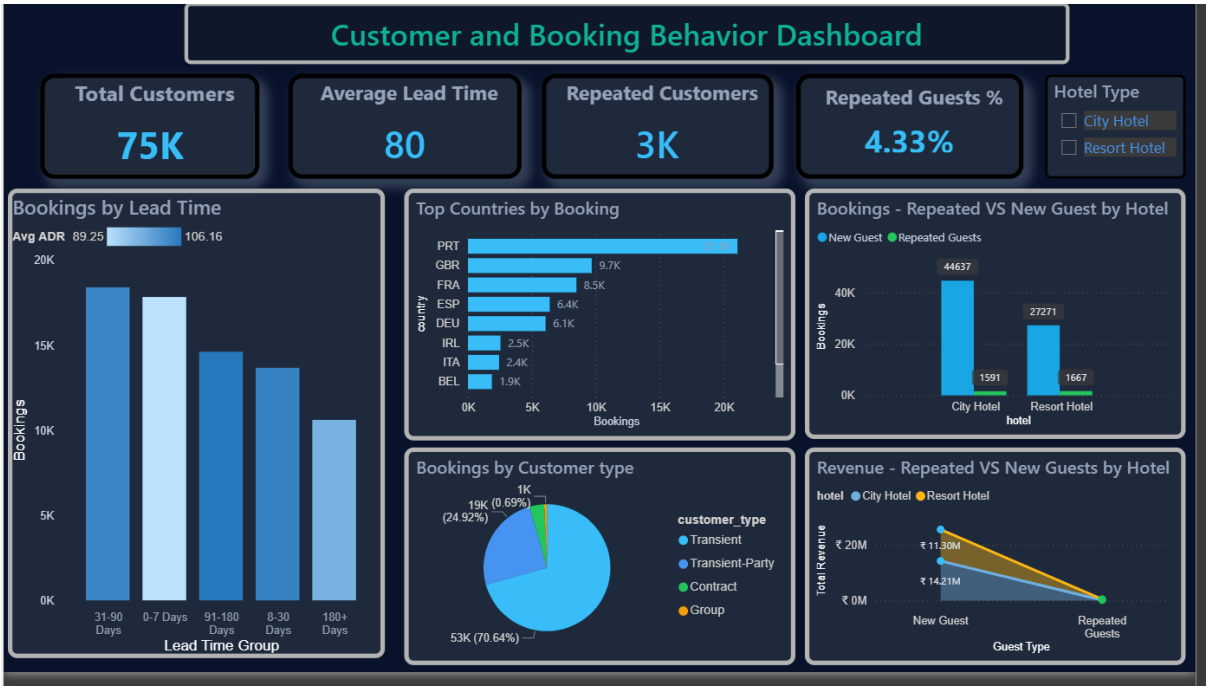
7.2 Customer and Booking Behaviour Dashboard

This dashboard focuses on understanding who the customers are, how they book, and how loyal they are. It enables targeted marketing and customer relationship management strategies.

Key Visualizations

- **Bookings and Avg ADR by Lead Time Group:** Bar chart showing the 31–90 day lead time group has the highest bookings (~18K), with average ADR ranging from ₹89.25 to ₹106.16

- **Bookings – Repeated vs New Guest:** Grouped bar chart showing most of bookings are from new guests, with only 4.33% from repeated customers by city and resort hotel.
- **Revenue – Repeated vs New Guests:** stacked area chart represents revenue of new and repeated guests showing that new guests generate 11.3M high revenue then repeated guests.
- **Bookings by Country – Top 10:** Bar chart with Portugal (PRT) dominating at 20K+ bookings, followed by Great Britain (GBR) and France (FRA)
- **Bookings by Customer Type:** Pie chart showing Transient customers at 70.64% of bookings



Business Story & Insights

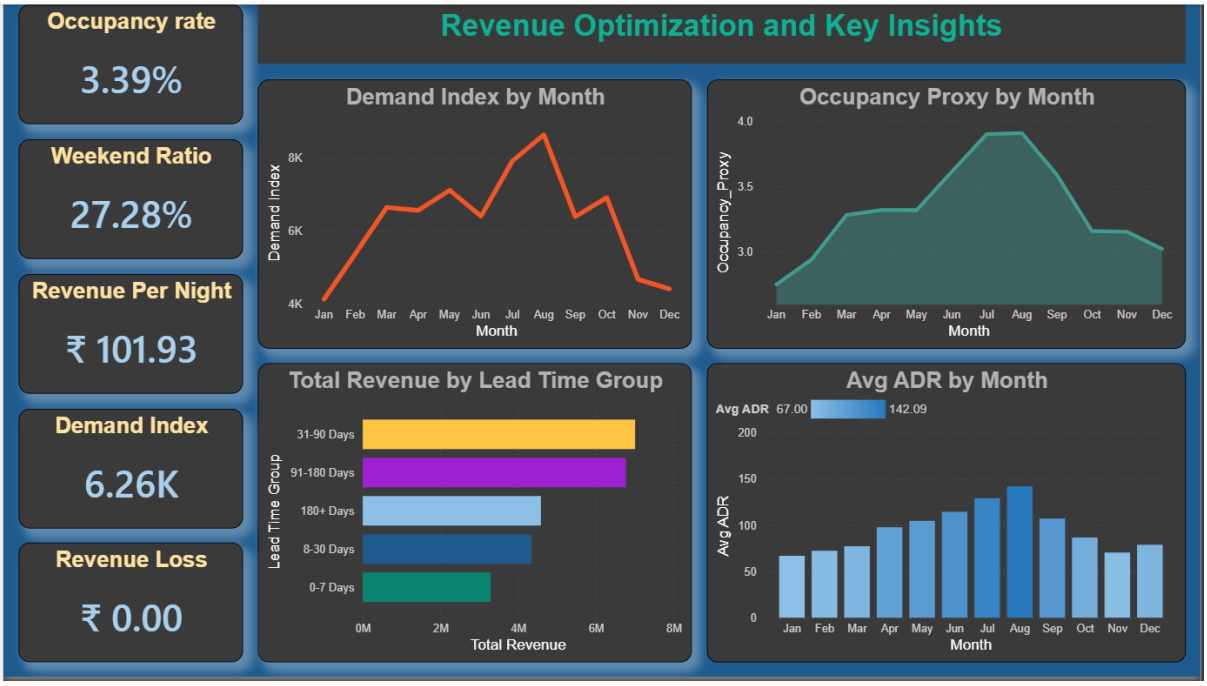
- The extremely low repeat guest rate (4.33%) signals a significant opportunity to improve customer retention through loyalty programmes and personalised post-stay communication
- Portugal is the dominant source market by a significant margin, suggesting targeted marketing campaigns in this geography could yield substantial bookings growth
- Bookings peak at 31–90 day lead times, indicating that early-bird pricing incentives in this window could maximise revenue capture
- Transient customers are the dominant booking type, reinforcing the need for flexible, individually-focused booking policies and pricing strategies

7.3 Revenue Optimization and Key Insights Dashboard

This dashboard focuses on strategic revenue optimization, combining occupancy analysis, demand indexing, and pricing performance to provide management with actionable levers for improvement.

Key Visualizations

- **Demand Index by Month:** Line chart showing demand peaking in August and experiencing a significant trough in December–January, confirming strong summer seasonality
- **Occupancy Proxy by Month:** Area chart tracking guest density per night across months, peaking in July–August at approximately 3.8
- **Total Revenue by Lead Time Group:** Horizontal bar chart showing the 31–90 day lead time group generating the highest revenue (~₹7M), followed closely by 91–180 day bookings
- **Avg ADR by Month:** Column chart revealing August achieves the highest ADR (~₹140), while January and November exhibit the lowest rates



Business Story & Insights

- The demand curve closely mirrors the ADR pattern — demand peaks in summer and rates are highest in August — confirming that the current pricing strategy is effectively capturing seasonal willingness-to-pay
- Revenue Loss showing ₹0.00 in the current view indicates that the filtered selection represents non-cancelled bookings — the clean revenue base for optimization analysis

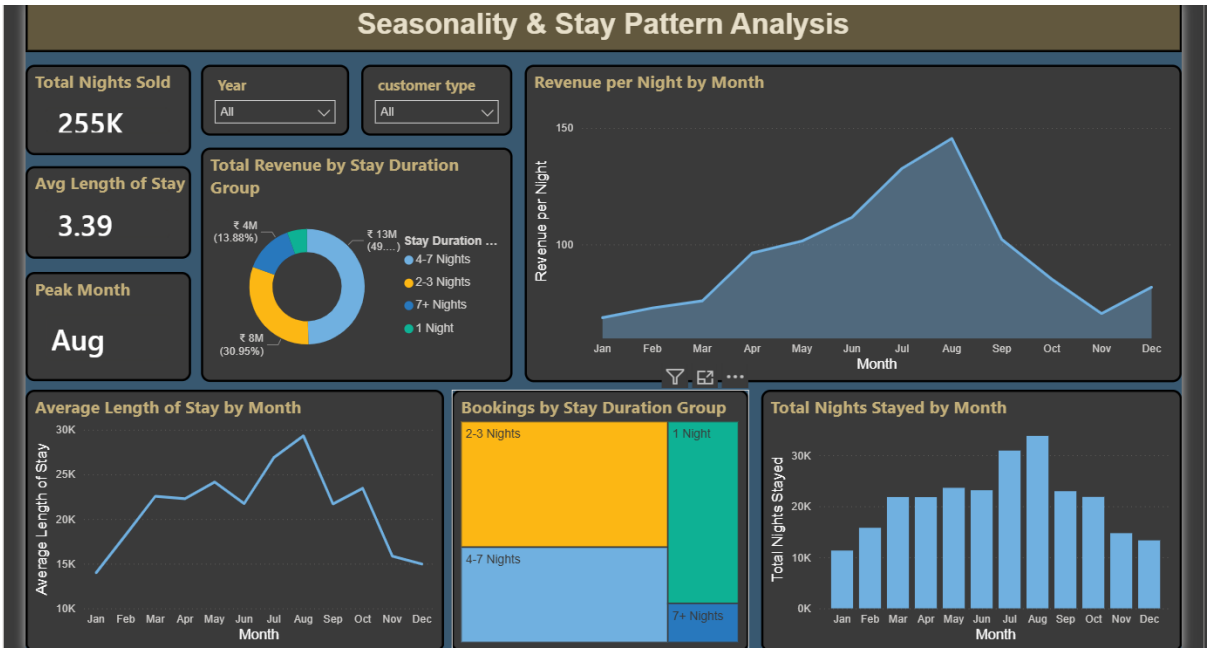
- The 31–90 day advance booking window generates the most revenue, suggesting hotels should incentivise bookings in this lead time range through targeted promotions
- Weekend ratio of 27.28% suggests significant weekday business, creating opportunities for weekday-specific corporate or extended-stay packages

7.4 Seasonality and Stay Pattern Analysis Dashboard

This dashboard investigates how booking patterns, stay durations, and revenue per night vary across months and seasons, providing insights for capacity planning and promotional calendar design.

Key Visualizations

- **Revenue per Night by Month:** Area chart showing a dramatic increase from ~₹65 in January to a peak of ~₹145 in August, followed by a sharp decline — the clearest illustration of seasonality
- **Total Revenue by Stay Duration Group**: Donut chart showing 4–7 night stays generating 49% of revenue (₹13M), followed by 2–3 nights at 31% (₹8M)
- **Bookings by Stay Duration Group:** Treemap visual showing 2–3 nights and 1 night are the most common stay durations, while 4–7 nights generate the most revenue despite lower booking counts
- **Average Length of Stay by Month:** Line chart showing sum average length of stay across month with slight peaks in mid-year months
- **Total Nights Stayed by Month:** Bar chart with August having the highest total nights (~30K), aligned with peak demand



Business Story & Insights

- August is unambiguously the peak month across all seasonality metrics — highest revenue per night, most nights sold, and highest demand index — creating a clear target for premium pricing strategies
- 4–7 night stays are the highest revenue-generating segment despite not being the most frequent booking type, suggesting extended-stay packages and mid-week bundles could be highly effective
- Short stay dominance (1–3 nights) across booking counts indicates a weekend and short-break travel market that can be targeted with curated weekend packages
- The consistent average length of stay around 3.39 nights suggests limited variation in guest itineraries — an opportunity to develop upsell strategies that extend stays through activity packages or incentives

8. Key Findings & Business Insights

Across all four dashboards, the following consolidated findings emerge from the HotelRevAI analysis:

8.1 Revenue Performance

- Total revenue of ₹26M was generated across 75,000 bookings with an average ADR of approximately ₹100
- August consistently emerges as the peak revenue month, with monthly revenues approaching ₹5M
- Online Travel Agencies (TA/TO) are the single largest revenue-generating distribution channel at ₹20M+
- Transient customers account for 74.37% of total revenue, making this segment the primary revenue driver

8.2 Customer Behaviour

- A very low repeat guest rate of 4.33% highlights a significant retention challenge and opportunity for loyalty programme investment
- Portugal (PRT) is the dominant source market with over 20,000 bookings, followed by Great Britain and France
- The 31–90 day advance booking window represents both the highest booking volume and the highest revenue contribution
- New guests generate 98.14% of total revenue, confirming that customer acquisition is currently the primary revenue mechanism

8.3 Seasonality & Stay Patterns

- Revenue per night follows a clear seasonal curve, rising from ~₹65 in January to ~₹145 in August
- 4–7 night stays generate nearly half of all revenue (49%), despite shorter stays (1–3 nights) being the most frequent booking type
- Total nights sold reached 255,000 across the dataset, with an average length of stay of 3.39 nights
- Weekend bookings represent 27.28% of total nights — a meaningful weekend leisure demand segment

9. Conclusion

The HotelRevAI project successfully demonstrates the application of Power BI as an end-to-end data analytics platform for hotel revenue intelligence. Beginning with structured data loading and preprocessing, through relational data modelling and a comprehensive DAX measure library, to the development of four interactive dashboards — the project delivers a complete analytical solution that transforms raw booking data into actionable business insights.

The four dashboards collectively tell a coherent business story: Online Travel Agency channels and Transient customers drive the majority of revenue; August is the consistent peak season and the most critical period for dynamic pricing; the repeat guest rate represents an untapped loyalty opportunity; and 4–7 night stays contribute disproportionately to revenue despite shorter stays dominating booking counts.

The Business Matrix framework established in this project ensures that every analytical metric is tied to a concrete strategic decision — whether that is optimising pricing by season, improving cancellation policies, developing loyalty programmes, or expanding direct booking channels. Together, these deliverables illustrate how data-driven intelligence can meaningfully improve operational efficiency and revenue performance in the hospitality industry.

Future Enhancements

- Integration of predictive analytics and machine learning models for demand forecasting and price optimisation
- Real-time data pipeline connectivity to replace static CSV imports with live hotel booking feeds
- Customer segmentation modelling using clustering algorithms to identify high-value guest profiles

- Competitor benchmarking layer with external market data to contextualise ADR performance against local market rates
- Automated anomaly detection alerts for unusual cancellation spikes or revenue drops